Example 4.15

Solve the differential equation $y^2dx+(xy+x^2)dy=0$

Solution

$$y^2dx+(xy+x^2)\,dy = 0$$

$$(xy+x^2)\,dy = -y^2dx$$

$$\frac{dy}{dx} = \frac{-y^2}{xy+x^2} \qquad (1)$$

It is a homogeneous differential equation, same degree in x and y

Put $y=vx$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$

$\therefore$ (1) becomes

$$v+x\frac{dv}{dx} = \frac{-v^2x^2}{x\ vx+x^2}$$

$$= \frac{-v^2}{v+1}$$

$$x\frac{dv}{dx} = \frac{-v^2}{v+1}-v$$

$$= \frac{-v^2-v^2-v}{v+1}$$

$$x\frac{dv}{dx} = \frac{-\left(v+2v^2\right)}{1+v}$$

Now, separating the variables

$$\frac{1+v}{v\left(1+2v\right)}\,dv = \frac{-dx}{x}$$

$$\frac{\left(1+2v\right)-v}{v\left(1+2v\right)}\,dv = \frac{-dx}{x}$$

$\left(\because\ 1+v=\ 1+2v-v\right)$

$$\frac{1}{v}-\frac{1}{1+2v}\,dv = \frac{-dx}{x}$$

On Integration we have

$$\int\left(\frac{1}{v}-\frac{1}{1+2v}\right)dv = -\int\frac{dx}{x}$$

$$\log v-\frac{1}{2}\log\left(1+2v\right) = -\log x+\log c$$

$$\log\left(\frac{v}{\sqrt{1+2v}}\right) = \log\left(\frac{c}{x}\right)$$

$$\frac{v}{\sqrt{1+2v}} = \frac{c}{x}$$

Replace v $= \frac{y}{x}$ we get

$$\frac{\frac{y}{x}}{\sqrt{1+\frac{2y}{x}}} = \frac{c}{x}$$

$$\frac{y\sqrt{x}}{\sqrt{x+2y}} = c$$

$$\frac{y^2x}{x+2y} = k$$

where $k = c^2$

Note

$\int\frac{1+v}{v^2+2v}\,dv$ can be done by the method of partial fraction also.

Example 4.16

Solve the differential equation $\frac{dy}{dx}=\frac{x-y}{x+y}$.

Solution:

$$\frac{dy}{dx}=\frac{x-y}{x+y} \qquad (1)$$

This is a homogeneous differential equation.

Now put $y=vx$ and $\frac{dy}{dx}=v+x\frac{dv}{dx}$

$\therefore$ (1) $\Rightarrow$ $v+x\frac{dv}{dx} = \frac{x-vx}{x+vx}$

$$= \frac{1-v}{1+v}$$

$$x\frac{dv}{dx} = \frac{1-v}{1+v}-v$$

$$= \frac{1-2v-v^2}{1+v}$$

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